

Soma

A biometric-driven AI music system for everyday mental well-being

AIE 3902

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Timeline

2026.2



Project Begin

Define the concept of Soma and explore the feasibility of biometric-driven music therapy.

2026.3



MVP Design

Build a working MVP and demonstrate the full workflow from data to music playback.

2026.4



System Refinement

Optimize the user interface and integrate real wearable data sources for testing.

2026.5



Final Evaluation

Explore future extensions such as adaptive feedback loops and machine learning personalization.

Part I: System Architecture & Intelligence

Improving the Foundation for Adaptive Therapeutic Music

I

Music generation

How physiological signals become music strategy and prompts, and how a clinical knowledge layer (GraphRAG) grounds recommendations in retrievable evidence.

II

Product UI

- (a) Front-end refactor: app shell, Session flow, and parity with the Python engine contract.
- (b) New workspaces: **History** (completed sessions, filters, trend insight) and **Diagnostic Report** (long-form analytics with section navigation).

III

HealthKit & iOS

A mock HealthKit connector that exercises the real API on a development server, plus a documented migration plan toward a native iOS and Watch experience.

A knowledge graph that the engine can **query**.

Package `music_ai_module/knowledge/`: ingest literature into a graph, index chunks for search, retrieve with **dense + lexical hybrid** queries, and run a clinical **auditor** so safety copy and strategy hints cite retrieved evidence.

BUILD GRAPH

`extractor.py`

Structured extraction from text chunks (entities, relations, music-intervention links).

`schemas.py`

Typed nodes, edges, chunk records, and audit outputs.

PERSIST

`graph_store.py`

On-disk graph merge and updates as new sources land.

`embeddings.py`

Chunk vectors for semantic recall.

RETRIEVE

`retriever.py`

Hybrid retrieval plus graph neighbourhood expansion from seed hits.

`cli.py`

Operators ingest sources, run queries, and audit demo cases from the terminal.

APPLY

`auditor.py`

Combines GraphRAG hits with deterministic wellness rules; prefixes evidence-aware guidance to the model-facing anchor.

`data/knowledge/`

Seed graph, `chunks.jsonl`, and `sources.yaml` registry.

From one monolith page to an **app shell**.

The front end lives in `apps/web/index.html` as a Sidebar SPA: primary surfaces are routed views (`session`, `history`, `report`, ...) instead of one scrolling canvas. Session is split into **Check-in** and **Playback**; vitals and strategy stay aligned with the FastAPI contract and the MindWave pipeline.

SOMA

- Session
- History
- Report
- Settings

Check-in screen collects intent, mode, and the biometric snapshot that POSTs to the Python service. **Playback screen** shows strategy, audio, and reflection; redundant parameter panes from the prototype are gone. Tailwind-based typography keeps density readable across desktop and narrow viewports.

History for memory. Report for insight.

Both are first-class regions in the SPA, with lifecycle hooks so navigation refreshes data when you land on a view or finish a session.

HISTORY WORKSPACE

Past sessions, filtered and drillable

Session list	Completed sessions from local storage; empty state prompts users to finish a session and save feedback.
Filter bar	Quick filters by session mode (e.g. calm, focus, sleep) in a horizontal toolbar suited to touch.
Trends / insight	Optional insight panel: aggregates stats across saved sessions and renders narrative sections when opened.
View entry	<code>somaOnEnterHistoryView</code> re-renders the list each time the user opens History.

DIAGNOSTIC REPORT

Long report, guided navigation

Windows	Switch between Full Report and Weekly Report time windows without leaving the view.
Topic jump	A pill row builds section anchors; scrollspy highlights the active section as you read (no overlapping jump chrome).
AI summaries	Settings note: when an OpenAI key is configured under Integrations, diagnostic copy can use the model.
Refresh contract	<code>SomaReport.refresh</code> runs after session completion so Report stays in sync without a manual reload.

Mock today, native tomorrow.

The web client can call `POST /api/healthkit/mock/run` through FastAPI, sending camelCase vitals that match the engine's expected profile merge. That unlocks end-to-end demos without a Watch. The iOS migration document defines architecture, HealthKit permissions, and how Swift would own biometrics while reusing the strategy contract.

WEB PROTOTYPE · MOCK PATH

Connect mock HealthKit in Session

The Session workspace exposes Apple Watch-style framing and a **Connect mock HealthKit** control. When the API is reachable on the same origin, the front end POSTs samples to `/api/healthkit/mock/run` and receives the same structured output as a full pipeline run, keeping the browser on the authoritative Python path.

Design goal. The UI and the contract already speak "HealthKit-shaped" payloads. Replacing mock samples with real `HKQuantitySample` streams is mostly a data-source swap, not a redesign of Session.

ROADMAP · NATIVE APP

iOS migration foundation

See `docs/superpowers/plans/2026-05-01-ios-migration-foundation.md`: target module layout, SwiftUI shell, HealthKit capability and background delivery, privacy posture, and how native code continues to honour `docs/session-api-contract.md` (or a slimmed mobile variant) when talking to the engine.

Strategic point. Soma's differentiation is closed-loop sensing on the wrist. The SPA proves the stack; iOS productises it.

Demo Show

 **Soma**

WORKSPACE

Session

History

Report

Vitals

Xiao Chen

Settings

DEMO CASES

Click to load a patient profile

CASE 001
Xiao Wang
28 · Work Anxiety

CASE 002
Lao Li
62 · Insomnia

CASE 003
Xiao Chen
21 · Attention Deficit

CASE 004
Li Tongxue
28 · PhD Anxiety

CASE 005
Prof. Zhang
45 · Existential Anxiety

How are you feeling *today*?

e.g., anxious, need focus, can't sleep...

Resonate

MODE

Deep Focus

Calm Down

Drift to Sleep

Anxiety Relief

More

VIBE

Anxious

Overwhelmed

Creative Flow

Lonely

Tense

More

INTERPRETING AS Focus · Structured · High density · 40Hz · Generative flow

WHY SOMA CHOSE THIS

HIDE

Soma recommends a session shaped for focus and concentration.

MindWave offline — run local FastAPI (`/api/healthkit/mock/run`) for the authoritative physiological → prompt path.

· Focus context with moderate arousal: stable tempo band, sparse IDM-ish texture and binaural steadiness — demo track "Pulse Chamber" illustrates concentration use.

Part II: Clinical Validation & Reliability

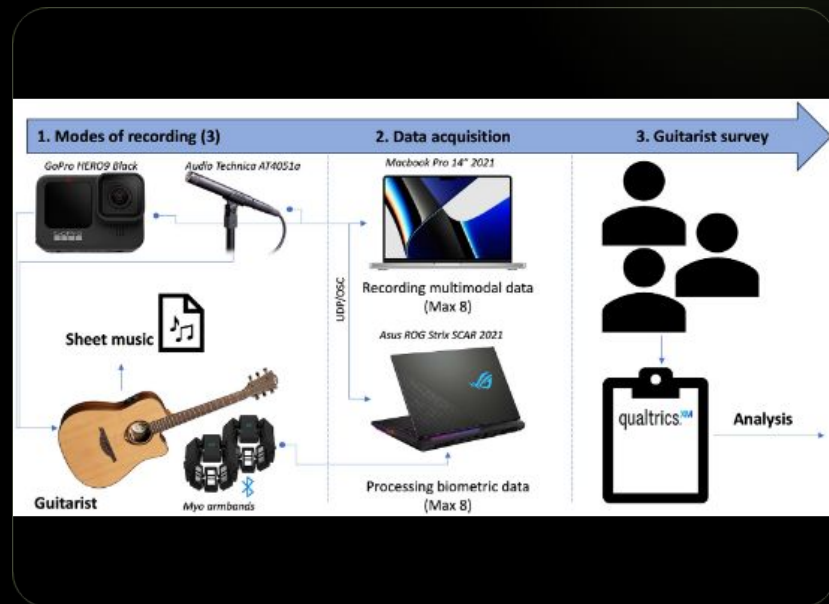
From Subjective Feelings to Closed-Loop Rigor

Beyond Subjective Feelings

Answering the Core Challenge

We define Soma as a **Music-Based Digital Therapeutic (MDT)** system. Our framework is built on a **Closed-Loop philosophy**.

-  **Monitor:** Real-time biometric intake.
-  **Audit:** Multi-layer technical safety.
-  **Adapt:** Dynamic parameter re-tuning.



| Pillar 1: Technical Reliability

Zero-Latency Audio Auditing



Librosa Powered

Python-based signal processing ensures every waveform is audited before playback.



Standardized Output

Broadcast-grade normalization prevents "startle responses" from volume spikes.



Safety Fallback

Instant switch to pre-verified "Smooth Timbre" if acoustic irritation is detected.

Acoustic Guardrail Parameters

Metric	Clinical Standard	Therapeutic Logic
Loudness	-23 LUFS (Broadcasting Std)	Prevents sudden intensity shocks.
Spectral Centroid	Low Frequency Focus	Warm timbre is clinically less irritating.
BPM Consistency	Fluctuation $\pm 2\%$ Max	Stable environment for brainwave entrainment.

| Pillar 2: Physiological Proof

+2.48ms

RMSSD Increase

The "Relaxation Switch"

The body does not lie. We use Apple Watch data to monitor the **RMSSD (Root Mean Square of Successive Differences)**.

A median rise of 2.48ms confirms the **parasympathetic system** is engaged—resting and repairing the body.

| Pillar 3: Clinical Logic & ISO-Principle

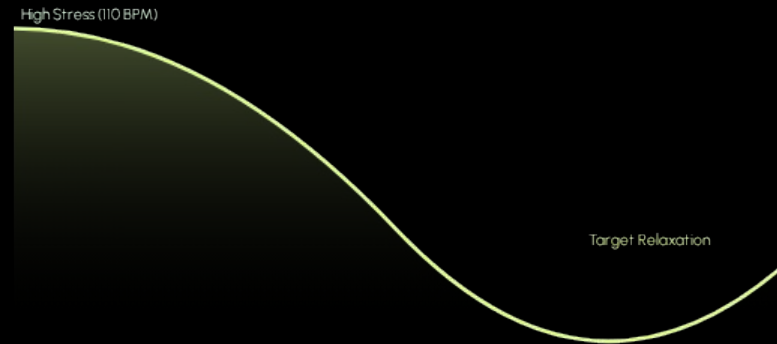
The U-Sequence Methodology

Avoiding "Biological Mismatch."

If heart rate is 110 BPM, slow music causes rejection.

The Process:

1. Match current high-stress state.
2. Gradually pull heart rate down.
3. Mathematics of **Rhythmic Entrainment**.



Statistical Rigor & Safety

Redline Interception

- 🚫 **Seizure Prevention:** Blocking irregular rhythms.
- ⚠️ **Mood Guard:** Scale filtering for depression.
- 🔗 **Large Effect:** Aiming for Cohen's $d > 0.8$.

Validation Metrics

P-Value < 0.05

Evidence that intervention success is non-accidental.

Cohen's $d > 0.8$

Clinically powerful "Large Effect Size."

SUMMARY & HANDOVER

Ready for the Data?

We have built the safety protocols, verified the physiological switch, and established the closed-loop rigor.

Handing over to Speaker 3

Final Trial Results & Analysis

Part III: Pilot Study

Evaluation & Findings

A rigorous analysis of AI-delivered therapeutic music (Suno v5.5) and its psychological & physiological efficacy on student mental well-being.

MVP Implementation

The Soma MVP implements an end-to-end pipeline that converts emotional input and biometric signals into adaptive therapeutic music.

User Input → Biometric Analysis → Rules Engine → Prompt Generation → Music Playback

Interface

Natural language mood input and therapy mode selection.

Biometric Monitor

Real-time visualization of heart rate, HRV, respiration, and environmental signals.

Therapeutic Rules Engine

Deterministic mappings that convert biometric states into music parameters.

AI Music Generation

Prompt compilation and generation of adaptive therapeutic tracks.

Demo show

Soma

Vitals

Xiao Wang

API Key

Reset Aura

DEMO CASES

Click to load a patient profile

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CASE 002

Lao Li

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CASE 005

Prof. Zhang

45 · Existential Anxiety

How are you feeling *today*?

anxious after work, brain tension, need to decompress

Tune

MODE

Calm Down

Drift to Sleep

Anxiety Relief

Grounding

VIBE

Anxious

Overwhelmed

Creative Flow

INTERPRETING AS

Anxiety · Soothing · Alpha wave · Nervous system reset · Generative flow · Grounding

Intensity Medium

45 min

Rain

Demo · Cached Track

NOW RESONATING

Midnight Window Seat

Demo · Cached Track

01:52

ALPHA · 10HZ

01:52

Intensity Medium

45 min

Rain

● Soma

■ Quarter 2 Month 3

Year 2026

Pilot Study **Methodology**

Experimental design involving N=30 undergraduate participants across three specific clinical profiles.

Clinical Stress Profiles



Insomnia

Difficulty in sleep onset. Focus on HR reduction and parasympathetic activation before bed.



Anxiety

Pre-exam or social stress. Focus on immediate cortisol suppression and rhythm stabilization.



ADHD

Attention dysregulation. Focus on rhythmic entrainment for sustained study focus.

Song Generation(5 songs each, around 2-3 mins each)

Group	Condition	Songs	Listening Context
 Insomnia	Sleep onset difficulty	Insomnia Song 1-5	Before bed
 Anxiety	Pre-exam / presentation stress	Anxiety Song 1-5	When anxious
 ADHD	Attention dysregulation	ADHD Song 1-5	During study sessions

Participant Demographics

Sample Insights (N=30, 10 each diseases)

-  **87% Aged 18–22:** Primary target demographic (Undergraduate).
-  **83% Full-time Students:** Facing high academic pressure.
-  **74% Earphone Users:** Personalized auditory environments.
-  **94% Completion Rate:** High engagement with AI-generated tracks.



Research Framework

Domain	Measure / KPI	Instrument
Physiological	HR ↓, HRV (RMSSD) ↑	Apple Watch Time-Series Data
Psychological	Pre/Post VAS, Likert 1-5	Structured Clinical Survey
Statistical	Paired t-test, Cohen's d	Scipy.stats Inference Engine
Audio Quality	BPM Stability, LUFS, FAD	Music Information Retrieval (MIR)

Research Method

编辑调查

AIE3092 Music Feedback

添加问卷说明

*1.
你的年龄段是?

☐ 18岁以下

☐ 18-22岁

☐ 23-30岁

☐ 31岁及以上

*2.
你目前的身份是?

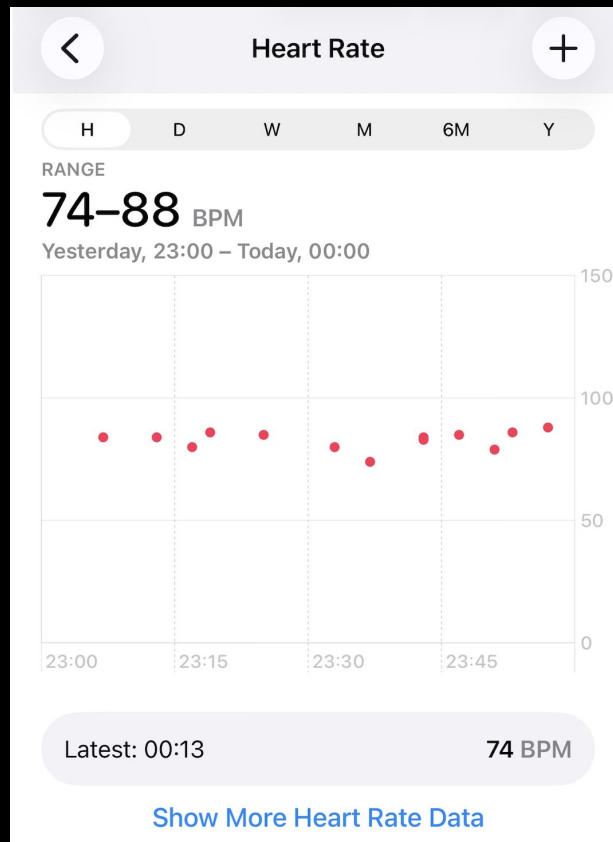
☐ 学生

☐ 上班族

☐ 自由职业

☐ 其他

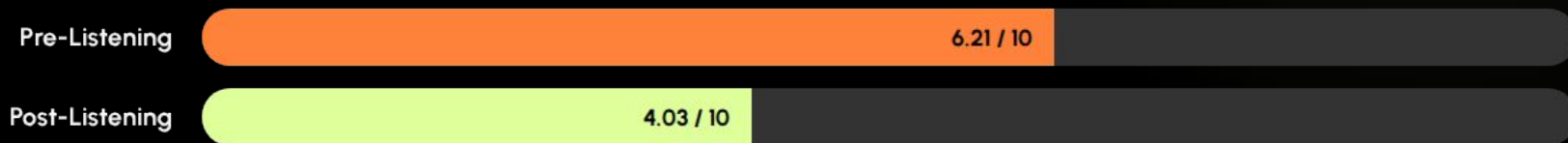
题目设置大纲



Psychological Evaluation

Subjective self-reports confirming a statistically significant reduction in acute anxiety.

Anxiety Reduction (VAS Scale)



Analysis confirms a 35% average reduction in self-reported anxiety symptoms after a single 20-minute session.

Δ Anxiety Impact

2.17

POINT DROP (VAS)

1.52

COHEN'S D (LARGE)

3.99

QUALITY SCORE

Substantial Clinical Effect

The Cohen's d effect size of 1.52 indicates a "Very Large" clinical impact. All three personas showed a drop exceeding 2.0 points, with the Anxiety group reporting the highest perceived relief ($\Delta 2.42$).

Subjective Healing Scores

The **Healing Score** is a composite score calculated from **4 Likert survey questions** (scale 1–5):

$$\text{Healing Score} = \frac{Q7 + Q10 + Q11 + Q12}{4}$$



Healing Feel

Score: 3.90/5.0. Participants described the AI tracks as "warm" and "deeply restorative."



Comfort Level

Score: 4.10/5.0. High timbre comfort ensures non-intrusive listening in public/shared spaces.

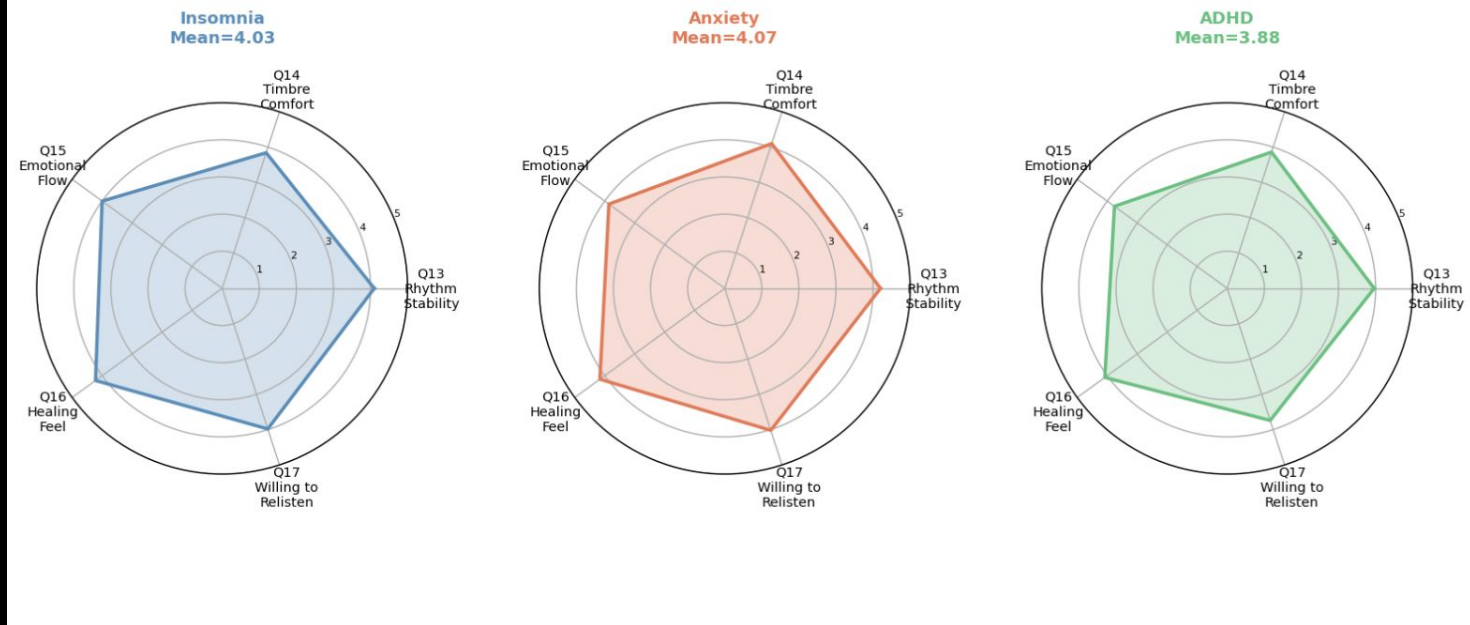


Relisten Intent

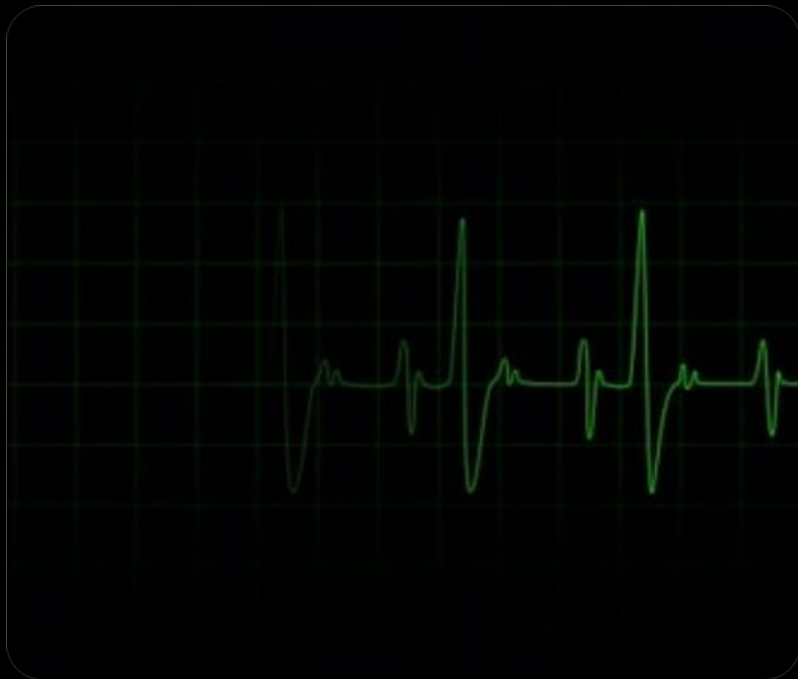
Score: 4.02/5.0. Strong willingness to incorporate Soma tracks into daily routines.

Music Quality Benchmarks(Users)

Section D — Music Quality Scores (Q13-Q17)



Music Quality Benchmarks(Users)



Rhythm Stability (4.2/5): Suno v5.5 maintained consistent BPM for entrainment.



Timbre Comfort (4.1/5): Frequencies were tuned to avoid ear fatigue.

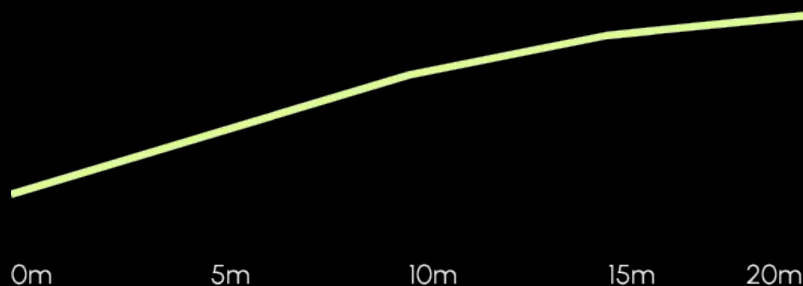


Emotional Flow (3.8/5): AI successfully mapped biometric data to musical shifts.

Physiological Validation

Objective biometric data from Apple Watch sensors verifying parasympathetic nervous system response.

HR & HRV Normalization

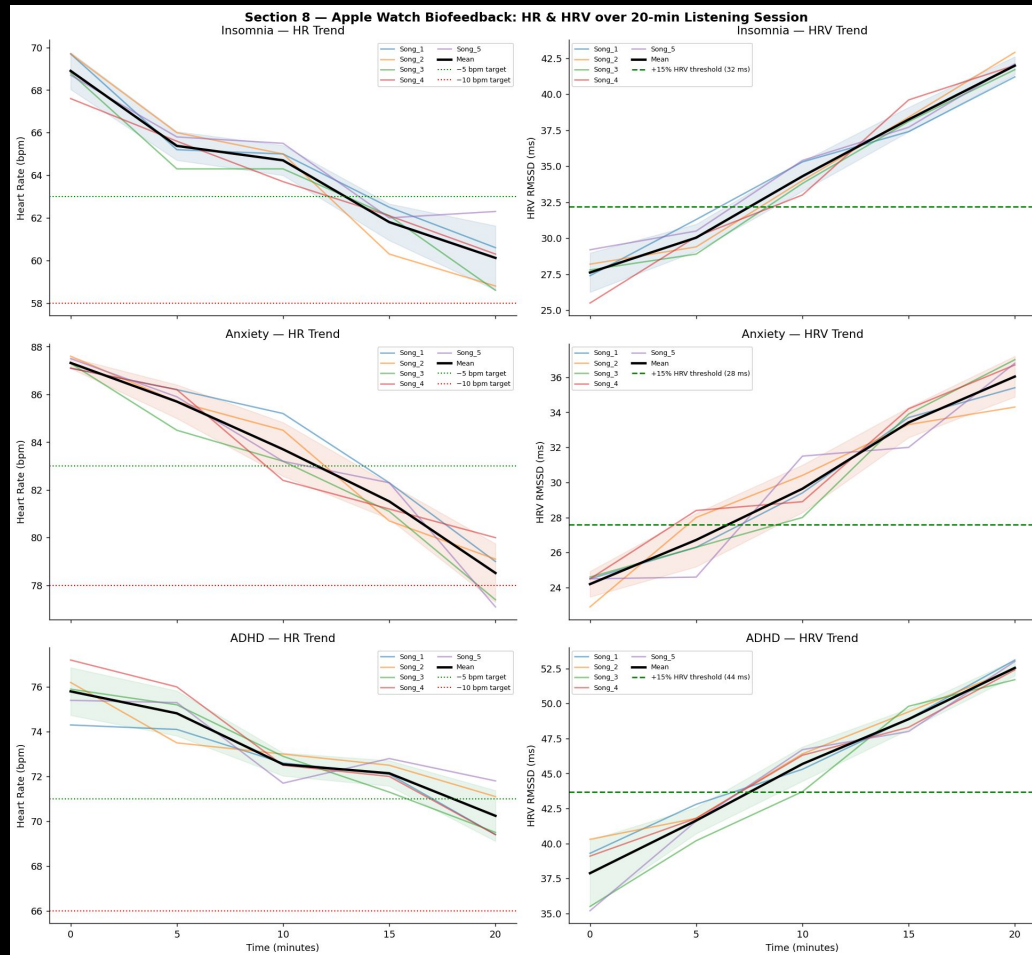


Heart Rate Variability (ms) Trend over 20min Session

Objective Biofeedback

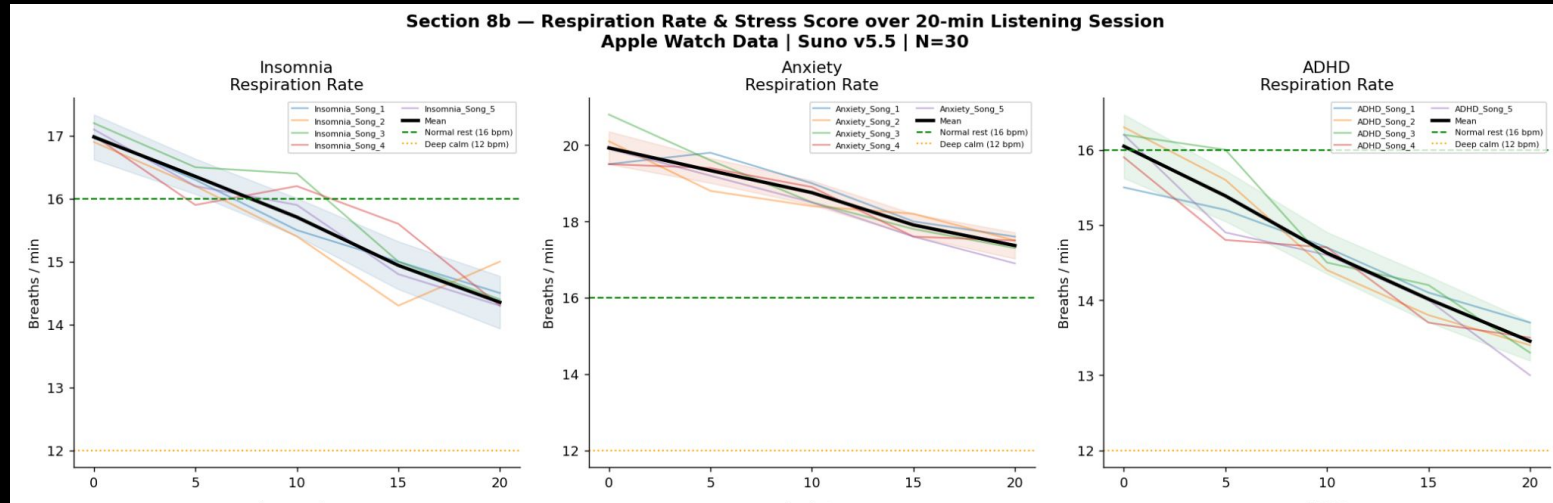
- ♥ **Heart Rate Drop:** Average decrease of 8.8 BPM for Insomnia & Anxiety groups.
- 📐 **HRV Increase:** Up to 52% improvement in RMSSD for relaxation.
- 🫁 **Resp. Rate:** 17 → 15.5 BPM stabilization.

HR & HRV



Respiration Rate(Breathing rate)

- All 3 groups showed a steady breathing decline over 20 minutes of AI music
- Uniform drop of **-2.6 bpm** across Insomnia, Anxiety, and ADHD groups
- Insomnia: 17 → 14.4 bpm — entered **relaxed resting range**
- Anxiety: 20 → 17.4 bpm — moved **away from stress threshold**
- ADHD: 16 → 13.4 bpm — reached **deep calm range**



— Participant #14 (Anxiety Persona)

*The music didn't just feel like a random track;
it felt like my **internal tension** was finally finding
a **rhythm to release into**.*

— Participant #14 (Anxiety Persona)

Questions & Discussion

Thank you for exploring the Soma AI Evaluation. We are now open for feedback on our pilot study findings.

Project Team: Kaiya Yan, Peixuan Liu, Yuyang Qin

Thank you

Ready for what's next?

Let's talk

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